



Modeling the Determinants of Systolic Blood Pressure Using Multiple Linear Regression in R

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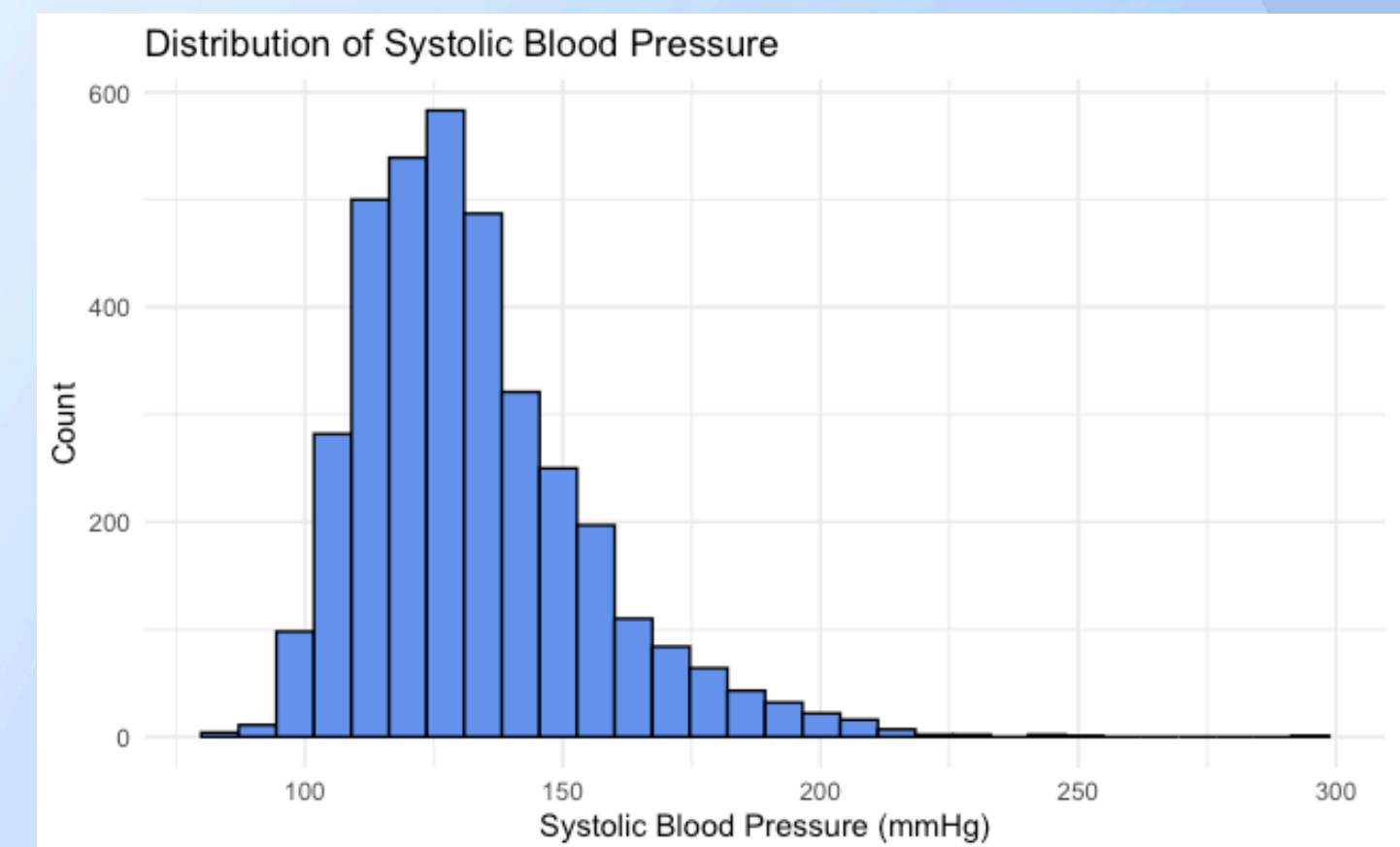
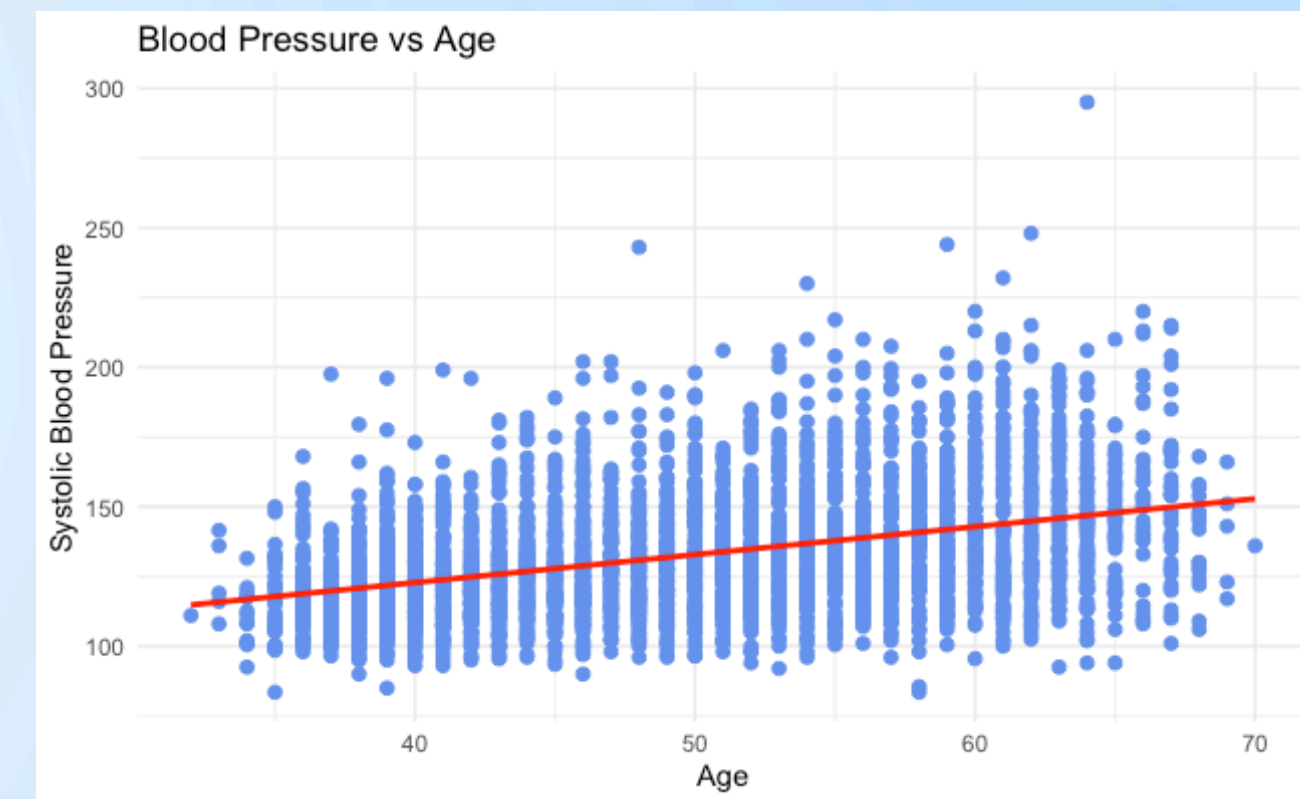
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Problem & Motivation

- Examine how influential the following factors are on systolic blood pressure:
 - Age
 - Sex
 - Blood pressure medication use
 - Cholesterol
- This problem can be applied well to statistical modeling with meaningful public health inquiries
- Goal:
 - Understand which factors are strongly related to higher systolic blood pressure
 - If relationships remain after accounting for other variables
 - Provides health insights and an opportunity to demonstrate the full regression workflow
- Ultimately, our analysis can provide a practical summary of how major health indicators are related to blood pressure

Data Description

- Utilized the **Framingham Heart Study** dataset
 - Source: NIH (National Heart, Lung, and Blood Institute)
 - At least 4210 observations
 - 16 columns
 - 15 attributes
- **Scatterplot:** Positive association between age and systolic blood pressure
- **Histogram:** Systolic blood pressure had a right-skewed distribution



Questions of Interest

1.

Which demographic and health-related factors are most strongly associated with higher systolic blood pressure in adults?

2.

After accounting for other relevant health factors, how are age, hypertension status, diastolic blood pressure, and related clinical measurements associated with systolic blood pressure?

Exploratory Data Analysis

- Identified and removed 645 missing values
 - final sample size: **3,658 observations**
- Explored/Cleaned dataset, response distribution, and relationships with predictors
 - 3,658 observations
- Found **right-skewed systolic blood pressure (sbp)**
 - distribution → suggested transformation
- Residual diagnostics from original model showed:
 - Non-normality in the **upper tail**
 - **Mild heteroscedasticity**
- Applied a **Box–Cox power transformation for response**:
 - Estimated $\lambda = -0.364$
 - Confidence interval indicated transformation was necessary
 - -0.51 through -0.2186
- Chose **log transformation** (instead of $\lambda \approx -0.5$) because:
 - Easier to interpret
 - Still significantly improved model diagnostics

Initial Model & Data Split

- Data were split into 70% training and 30% testing sets using `caret`

- Initial model fit:

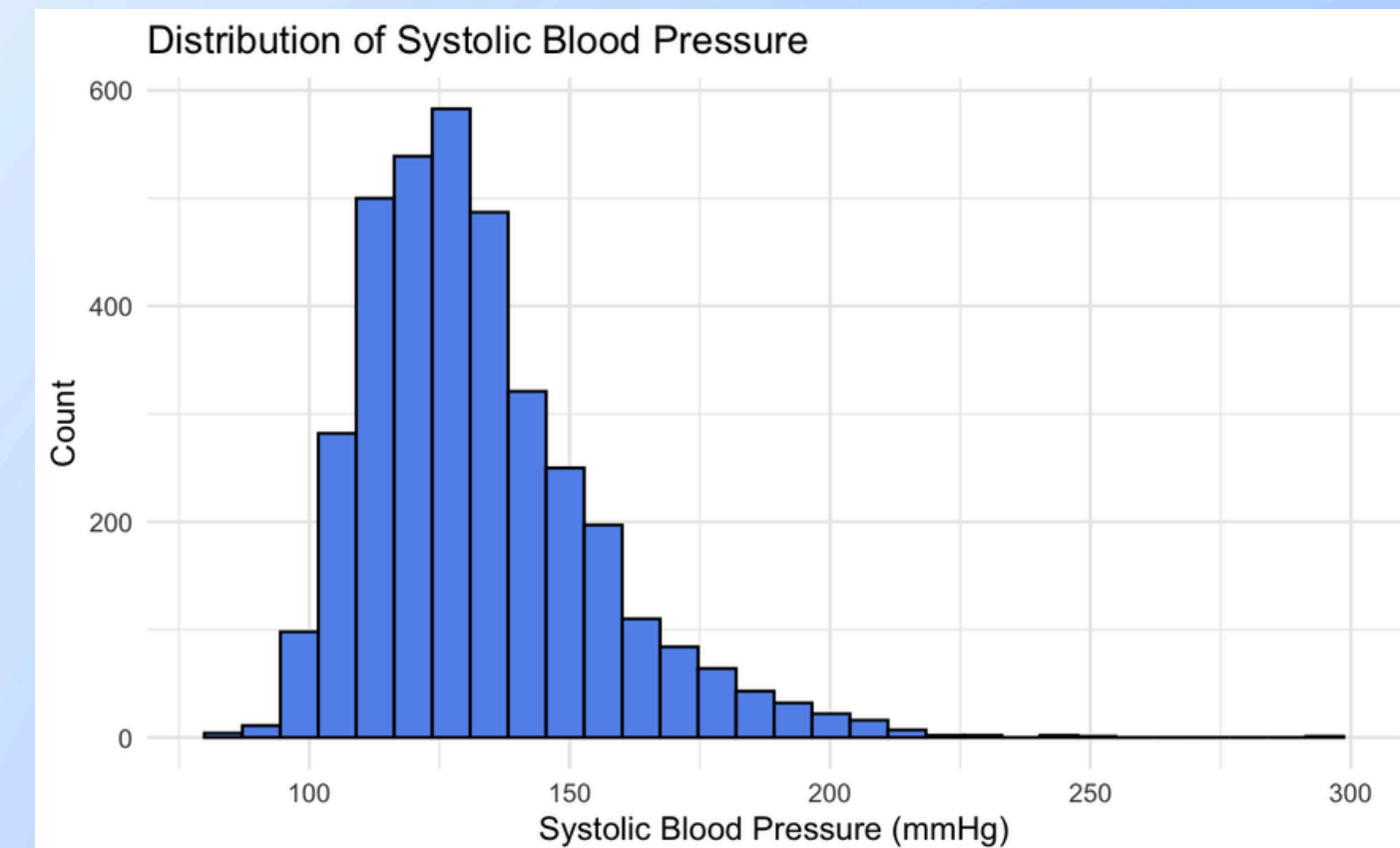
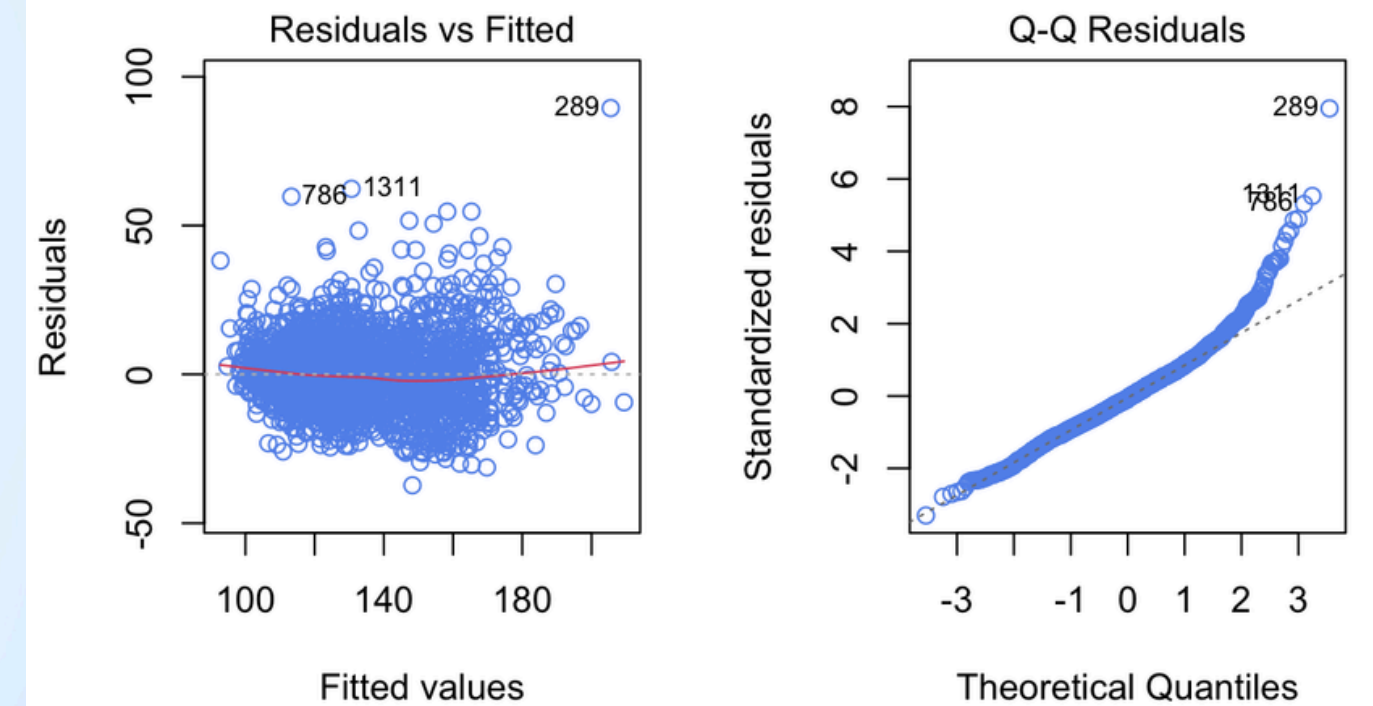
- $R^2 = 0.7371$
- $R^2_{adj} = 0.7355$

- MLR Model:

- $$\begin{aligned} sysBP = & \beta_0 + \beta_1(\text{male}) + \beta_2(\text{age}) + \beta_3(\text{education}_2) + \beta_4(\text{education}_3) + \beta_5(\text{education}_4) \\ & + \beta_6(\text{currentSmoker}) + \beta_7(\text{cigsPerDay}) + \beta_8(\text{BPMeds}) + \beta_9(\text{prevalentStroke}) + \beta_{10}(\text{prevalentHyp}) \\ & + \beta_{11}(\text{diabetes}) + \beta_{12}(\text{totChol}) + \beta_{13}(\text{diaBP}) + \beta_{14}(\text{BMI}) + \beta_{15}(\text{heartRate}) + \beta_{16}(\text{glucose}) + \beta_{17}(\text{TenYearCHD}) + \varepsilon \end{aligned}$$

Why We Transformed the Response?

- **Original model showed:**
 - Non-normal residuals
 - Mild heteroscedasticity
- **Shapiro-Wilk:** $w = 0.96612, p < 2.2e - 16$
- **ncvTest:** $Chisquare = 297.1593, Df = 1, p = < 2.22e - 16$
- **PowerTransform Estimate:** $EstPower : -0.3655$
 - We used $\log(\text{sysBP})$ for simplicity and interpretability



Final Model

Final Predictors retained:

- Male
- Age
- prevalentHyp
- totChol
- diaBP
- heartRate
- Glucose

Final Model (Reduced):

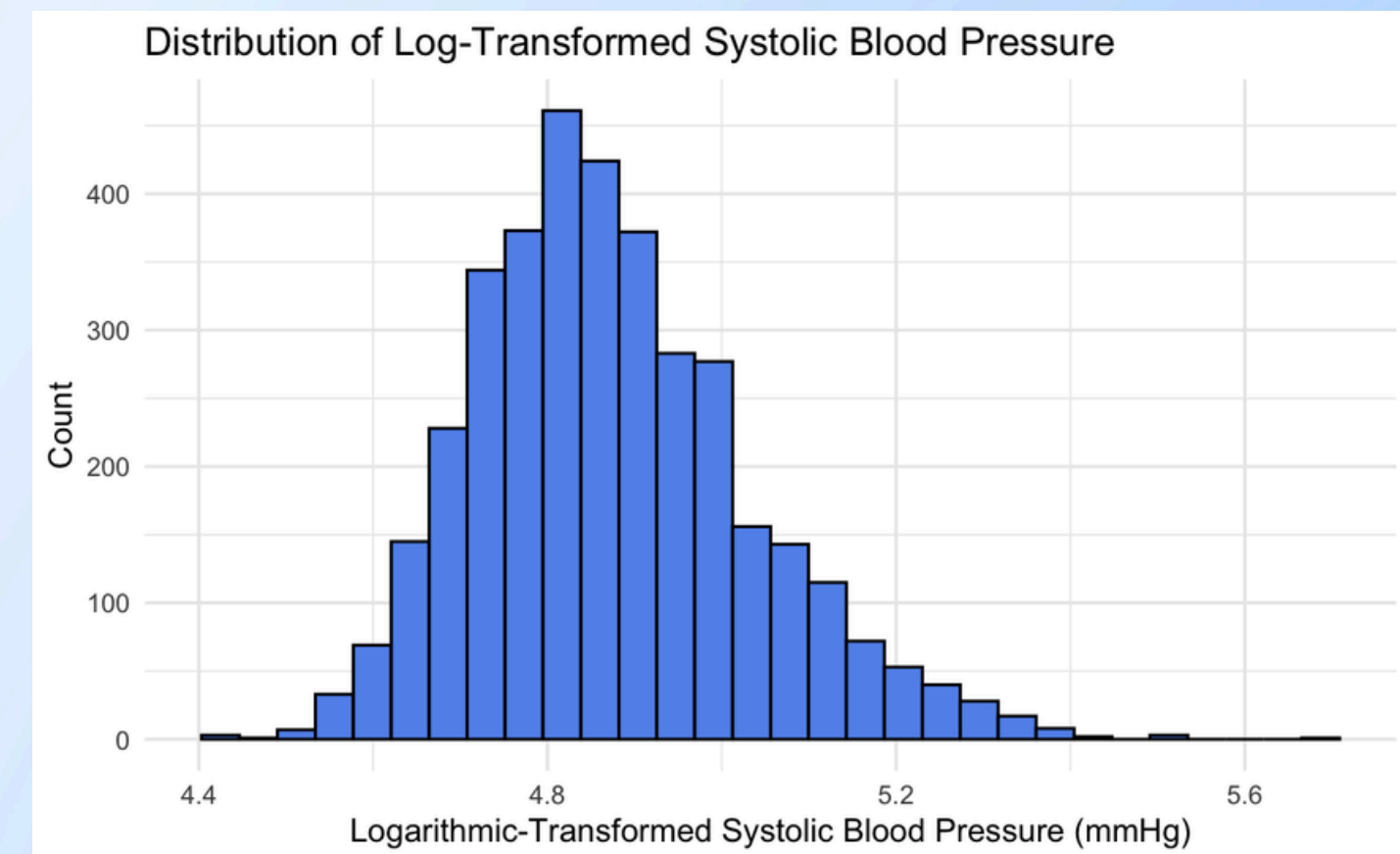
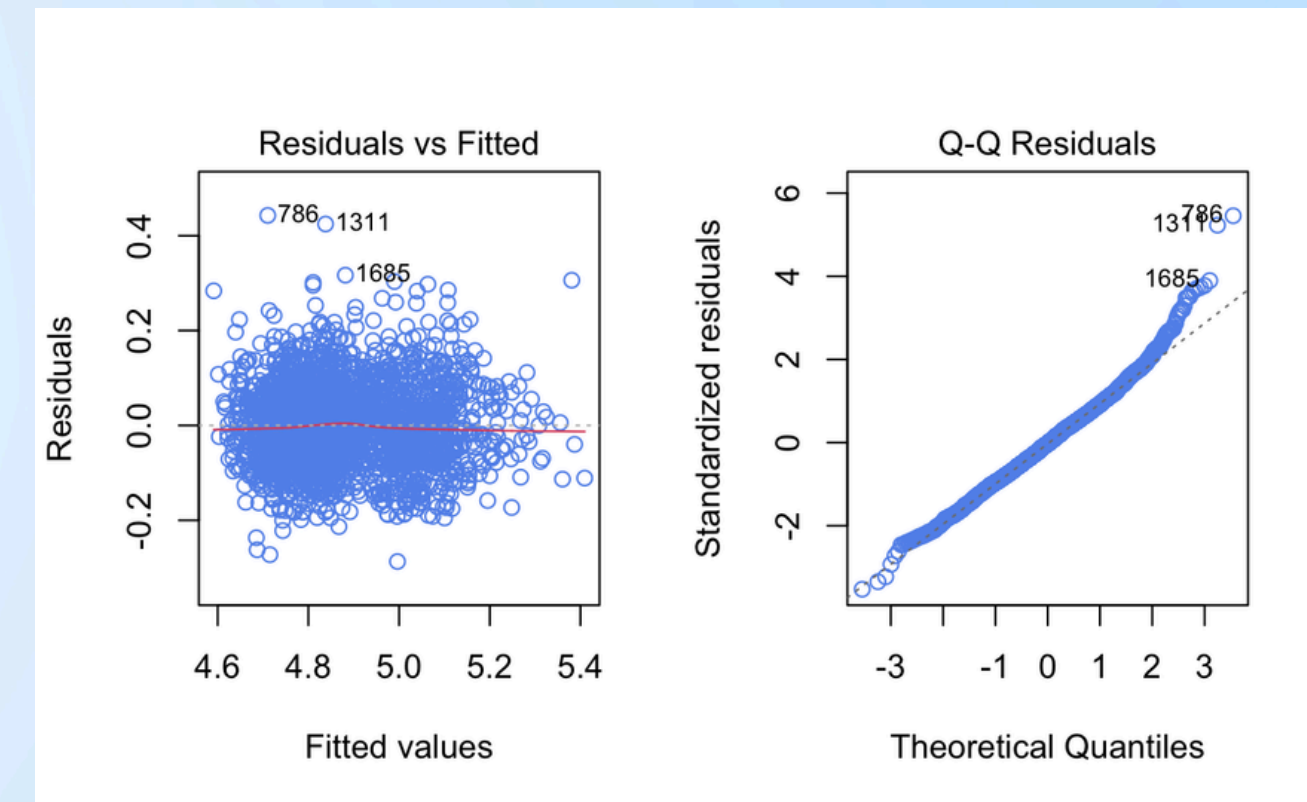
- $\log(\text{sysBP}) = 4.003 - 0.01650(\text{male}) + 0.003075(\text{age}) + 0.04096(\text{BPMeds}) + 0.09423(\text{prevalentHyp}) + 0.00008607(\text{totChol}) + 0.007450(\text{diaBP}) + 0.0003975(\text{heartRate}) + 0.0003266(\text{glucose})$

Bottom Line:

- Backward elimination used a removal criterion of 0.10

Results of Final MLR Model

- **Final model fit:**
 - $R^2 = 0.7348$
 - $R^2_{adj} = 0.734$
- Model explains ~73.4% of variation in systolic blood pressure
- **No serious multicollinearity:**
 - All VIF values close to 1
- Diagnostics improved after log transformation
- **Slight Q-Q plot deviation caused by 3 extreme residuals (786, 1311, 1685):**
 - High systolic BP with relatively low diastolic BP
 - Consistent with *Isolated Systolic Hypertension*
 - Valid observations and not influential (low leverage)

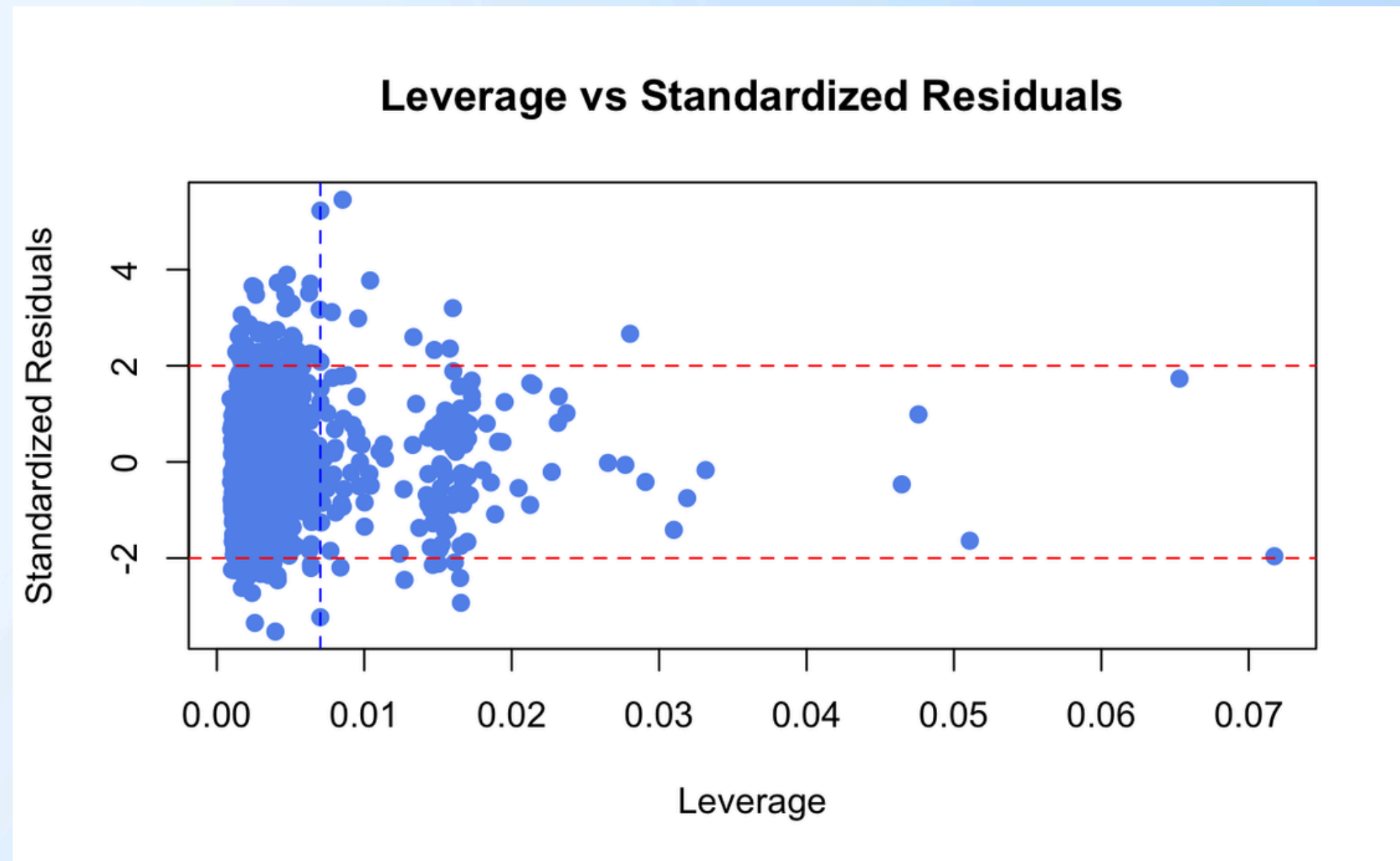


Interpretation of Key Predictors

- Age: ~0.31% increase in systolic BP per year
- Diastolic BP: ~0.75% increase (strongest predictor)
- Prevalent hypertension: ~9.4% higher systolic BP
- Blood pressure medication: ~4.1% higher systolic BP
- Male: ~1.65% lower systolic BP
- Total cholesterol: ~0.009% increase per unit
- Heart rate: ~0.040% increase per beat
- Glucose: ~0.033% increase per unit

Leverage Table

- High Leverage Points are between 0.05 and 0.07.
- Most of our points are within low leverage range.
- Our model is dominated by low leverage instead of high leverage.
- All non-influential points are past 0.05



Conclusion

- From this study we have confirmed that the strongest health indicators that connected to a person's systolic blood pressure was their diastolic blood pressure, age and cardiovascular status.

Limitations

- Slight deviation from normality in Q-Q plot (upper tail)
- A small number of extreme residuals identified (786, 1311, 1685)
- These correspond to **unusual but valid physiological cases**
 - High systolic blood pressure (past 150 mmHg) with relative low diastolic blood pressure
 - Known as *isolated systolic hypertension* (medically valid)
 - Not data errors and not influential observations (low leverage)